INCH-POUND
MIL-DTL-24643/4F
27 November 2023
SUPERSEDING
MIL-DTL-24643/4E
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 600 VOLTS, TYPE LSMCOS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Non-Watertight)

First - Copper conductor, coated or uncoated, class K (see table I for size).

Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.

Third - Thermoset insulation (see table I for wall thickness). Sizes 2, 4, and 7 shall have special identification code applied by method 3. Size 5 shall have one shielded pair (colors black and white) and three singles (colors red, green, and blue) applied by method 3. Size 6 shall have two shielded pairs (colors black and white for first pair and blue and red for second pair) and two singles (colors yellow and green) applied by method 3.

Fourth - (For all pairs only) An optional binder.

Fifth - (For all pairs only) Braided shield of 34 or 36 AWG coated or uncoated copper. Braid angle 30 to 35 degrees; minimum coverage of 88 percent.

Sixth - The required number of singles and shielded pair(s) to be cabled together with a left-hand lay. The length of lay (pairs and cables) shall be as required for the completed cable to meet the twisting and bending endurance tests; and for sizes 5 and 6, the additional mutual inductance test. Fillers may be used as necessary to form a firm, well-rounded assembly. A single conductor (or pair) shall not be used as a central core for the cable. The size 7 cable shall have a central core of fiberglass.

Seventh - An optional binder.

Eighth - Braided shield of 34 or 36 AWG coated or uncoated copper. Braid angle of 30 degrees minimum. Minimum coverage of 88 percent.

Ninth - An optional binder.

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Tenth - Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.040 inch. See table I for cable overall diameter.

NOTE 1: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

NOTE 2: Omit eighth and ninth construction steps for sizes 5 and 6.

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TABLE I. Details.

Military part no. M24643/4	Type and size	Conductors		Insulation thickness min. avg. (inch)	Overall diameter (inch)		Voltage withstand conductor to conductor (volts)	Conductor resistance maximum (ohms)	Insulation resistance minimum (megohms)	Roller diameter for bending endurance max. (inch)
		No.	AWG		min.	max.				
-01UO	LSMCOS-2	2	18	0.023	0.440	0.460	2,000	6.96	200	1.4
-02UO	LSMCOS-4	4	18	0.023	0.490	0.510	2,000	6.96	200	1.5
-03UN	LSMCOS-5	5	20	0.013	0.375	0.390	1,000	11.28	100	1.2
-04UN	LSMCOS-6	6	20	0.013	0.460	0.480	1,000	11.28	100	1.4
-05UO	LSMCOS-7	7	18	0.023	0.575	0.595	2,000	6.96	200	1.8

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REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	See table I
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	See table I
Conductor to shield (pair or overall)	500
Overall shield to ground (LSMCOS-2, LSMCOS-4, and LSMCOS-7 only)	1,500
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	See table I
Conductor to shield (pair or overall)	See table I
Conductor continuity	No failure
Shield continuity	No failure
Jacket flaws	No failure

Group A:

Visual and dimensional	No failure

Group B:

Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35

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Twisting and bending endurance (see table I for roller size)	
At 167 °F (75 °C) (cycles, min.)	6,000
At -22±3.6 °F (-30±2 °C) (cycles, min.)	2,000

Group C:

Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure
Mutual inductance (LSMCOS-5 and LSMCOS-6 only) (microhenries, max.)	0.0135
Shield (conformance to material, construction, and coverage)	No failure
Surface transfer impedance (types LSMCOS-2, LSMCOS-4, and LSMCOS-7 only)	
Milliohms per meter (max.)	Not applicable
Electromagnetic pulse (EMP) response (dB, min.)	Not applicable

Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18

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Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
All sizes	2,000

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-021)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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